Byeonggil Jun

Computer Engineering Ph.D. Student
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https://byeonggiljun.github.io/
Arizona State University

EDUCATION

Arizona State University — Tempe, AZ, USA
Ph.D. student in Computer Engineering — Jan.2024 – Present

- School of Comuting and Augmented Intelligence
- Advisor: Prof. Hokeun Kim
- Cumulative GPA: 3.97

Hanyang University — Seoul, South Korea
B.S. in Electronic Engineering — Mar.2018 – Feb.2024

- Department of Electronic Engineering
- Advisor: Professor Hokeun Kim
- Cumulative GPA: 4.26/4.5 (97.9/100)

AREAS OF INTEREST

Embedded Systems, Cyber-Physical Systems, Real-time Systems, Computer Architecture

PUBLICATIONS

Byeonggil Jun, Deeksha Prahlad, Mehdi Ghasemi, and Hokeun Kim, "Embodied AI Launcher: A Tool for Modeling, Simulation, and Deployment Evaluation of Transformer-Based Embodied AI Tasks at the Edge", Forum on specification & Design Languages (FDL), To appear, 2026

Hwisoo So, Byeonggil Jun, Chanhee Lee, Hokeun Kim, and Aviral Shrivastava, "PREFACE: Proactive Re-executions for Fault-aware Mixed-criticality Environments", Design, Automation and Test in Europe Conference (DATE), Verona, Italy, 2026

Peter Donovan, Erling Jellum, Byeonggil Jun, Hokeun Kim, Edward A. Lee, Shaokai Lin, Marten Lohstroh, and Anirudh Rengarajan, "Zero-Delay Cycles in Distributed Discrete-Event Systems using Lingua Franca", ACM Transactions on Modeling and Computer Simulation (TOMACS), Just Accepted, Oct 2025

Byeonggil Jun, Megan Kuo, Aditya A. Krishnan, and Hokeun Kim, "Towards Efficient Privacy-Preserving Federated Learning on Edge with Reconfigurable FPGA", Forum on specification & Design Languages (FDL), Schloß Rheinfels, St. Goar, Germany, Sep 2025

Byeonggil Jun, Edward A. Lee, Marten Lohstroh, and Hokeun Kim. 2025. Improving the Efficiency of Coordinating Timed Events in Distributed Systems. In Proceedings of the 39th ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (SIGSIM-PADS '25). Association for Computing Machinery, New York, NY, USA, Jun 2025

Byeonggil Jun, Edwared.A.Lee, Marten Lohstroh, and Hokeun Kim, "Efficient Coordination for Distributed Discrete-Event Systems", International Symposium on Formal Methods and Models for System Design (MEMOCODE), Raleigh, NC, Sep 2024.

Byeong-gil Jun, Dongha Kim, Marten Lohstroh, and Hokeun Kim, "Reliable Event Detection Using Time-Synchronized IoT Platforms," Time-Centric Reactive Software (TCRS), San Antonio, TX, May 2023.

HONORS AND AWARDS

ESWEEK 2024 Student Travle Grant, Sep 2024
CPS-IoT Week 2023 Travel Grant Award, May 2023
Academic Achievement Excellence Award, Hanyang University, 2022-2, 2022-1, 2021-1

RESEARCH EXPERIENCE

ASU KIM Lab — Tempe, AZ, USA
Research Assistant — Jan.2024 – Present

- Lead a Lingua Franca project on enhancing performance of HLA-based distributed discrete-event systems
- Lead a project on developing optimal network-on-chip interconnection for heterogeneous chiplet-based systems running complex AI workloads
- Participate in a project on privacy-preserving federated learning systems using homomorphic encryption
- Participate in a project on soft-error tolerant real-time systems

HYU IoT Lab Seoul, South Korea
Undergraduate Research Intern Mar.2022 – Dec.2023

- Participated in Lingua Franca projects on improving reactor models using C and TypeScript
- Led project on detecting events reliably using simple distance sensors with the TypeScript targeted reactor model

ACADEMIC SERVICES

Secondary Reviewer

- Design, Automation, and Test in Europe Conference and Exhibition (DATE 2026)
- 31st Asia and South Pacific Design Automation Conference (ASP-DAC 2026)
- Design, Automation, and Test in Europe Conference and Exhibition (DATE 2024)
- ACM SIGBED International Conference on Embedded Software (EMSOFT 2023)